

Does Online Mean-Reversion Beat Equal Weight?

A Reproduction and Pre-Registered Tests on US-Listed ETFs

Yelnil Gabo

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Abstract

Lahanis et al. (2025) report that mean-reversion algorithms for online portfolio selection multiply wealth hundreds to over a thousand times on NASDAQ-100 stocks over 1998–2009. We reproduce their tables exactly from their code and data, then test their variant of Confidence-Weighted Mean Reversion (CWMR) on 16 liquid US-listed ETFs over 1999–2024, deciding on each session’s close and trading at the next session’s open. Traded daily, it returns 12.1% a year at 0.7 basis points per dollar traded, against 8.4% for an equal-weight portfolio of the same ETFs and 8.1% for SPY, but it trades 170 times its account a year and its lead over equal weight is gone at 2.2 basis points. Learning from weekly prices gives the same return with less trading (break-even 5.2 basis points), and most of that lead disappears if each trade is made one session later. After correcting for the comparisons made, no variant’s Sharpe ratio is significantly higher than equal weight’s or SPY’s (smallest Holm-adjusted $p = 0.11$). Weekly and monthly variants passed a pre-registered test against SPY that equal weight also passes, and a monthly variant chosen on recent data failed its pre-registered test on 2025–2026. We find no statistically significant edge over holding the same ETFs.

1 Introduction

Online portfolio selection treats investing as a repeated game. Each period an algorithm holds a portfolio, observes the period’s price changes and updates its weights by a fixed rule. Cover (1991) showed that one such rule, the universal portfolio, achieves asymptotically the same growth rate as the best constant-rebalanced portfolio chosen in hindsight. A later family of algorithms bets on reversal instead. Anticor (Borodin et al., 2004), passive aggressive mean reversion (PAMR; Li et al., 2012), confidence weighted mean reversion (CWMR; Li et al., 2011, 2013), online moving average reversion (OLMAR; Li et al., 2015) and robust median reversion (RMR; Huang et al., 2013) move weight from recent winners to recent losers. The papers that introduced them report very large wealth multiples on a common set of benchmark datasets; Li and Hoi (2014) survey the family.

Lahanis et al. (2025), a publicly released working paper, implement about 25 of these algorithms and test them on NASDAQ-100 stocks; their data run from 1998 to the end of 2009, although the paper gives the sample as 1998–2010. Their best result, CWMR, multiplies wealth 1,590 times. They disclose that they ignore trading costs and that the period may not represent recent markets.

The question here is whether any version of their CWMR survives trading costs and timing on instruments a retail investor can buy today. The backtests use `btest`, the author’s backtesting software, and every table on the ETFs except the holdout is also computed by an independent replication from public data.

Related work. Trading costs have been studied in this literature from the start. Blum and Kalai (1999) extend universal portfolios to proportional costs, Li et al. (2011) report that CWMR withstands small costs on their benchmark datasets, Li et al. (2018) design an algorithm that trades reversion against costs, and Li et al. (2016) publish a toolbox for these experiments. Closest to this paper, Moon et al. (2019) test PAMR, OLMAR and cost-optimised variants on S&P 500 stocks over 2000–2017; they find that the standard benchmark datasets favour mean reversion and that the strategies may fail when explicit or implicit costs exist. At the level of assets, short-horizon reversal is documented for individual stocks at weekly (Lehmann, 1990) and monthly (Jegadeesh, 1990) horizons, while industry portfolios show momentum (Moskowitz and Grinblatt, 1999), so the stock-level evidence does not carry over directly to sector and index ETFs. Bid-ask bounce makes measured prices reverse (Roll, 1984), and part of measured short-term reversal in stocks is due to it (Kaul and Nimalendran, 1990; Conrad et al., 1997); the returns to reversal strategies can also be read as pay for supplying liquidity (Nagel, 2012). An equal-weight portfolio is a hard benchmark for optimised portfolios in general (DeMiguel et al., 2009).

Beyond Moon et al. (2019), this paper adds an exact reproduction of a recent paper, which uncovers defects in its code; a test on index and sector ETFs rather than stocks, with spreads measured from quotes and fills that could have been executed; equal weight in the same funds as the benchmark; and confirmatory tests specified before they were run, with the deviations from that plan reported. Section 2 describes the data and methods. Section 3 reproduces the original paper. Section 4 tests CWMR on the ETFs over 1999–2024, Section 5 asks what costs it can bear, Section 6 examines the weekly variant and Section 7 reports the holdout test. Section 8 reports the pre-registered tests and where we departed from their plans. Section 9 discusses what the evidence supports.

2 Data and methods

2.1 Algorithm

All tests use the authors’ CWMR as written in their public repository, ported to btest and checked against their function to within 7.1×10^{-9} per weight on their data. Their code describes it as a simplified version of the published algorithm (Li et al., 2011, 2013). Among other differences, its step size has no confidence term, a learning-rate factor η is added and the update is not mean-centred. We keep their parameters throughout ($\epsilon = 0.89$, $\theta = 0.92$, $\eta = 0.93$) and never re-tune them.

Each period the algorithm receives a vector x of price ratios, one per asset ($x_i = 1.02$ means asset i rose 2%). With current weights b and covariance Σ (initially the identity), it computes

$$\lambda = \eta \max\left(0, \frac{b^\top x - \epsilon}{x^\top \Sigma x}\right), \quad (1)$$

$$\mu = b - \lambda \Sigma x, \quad (2)$$

$$\Sigma^{-1} \leftarrow \Sigma^{-1} + 2\lambda\theta xx^\top, \quad (3)$$

$$b \leftarrow \Pi_\Delta(\mu), \quad (4)$$

where Π_Δ projects onto non-negative weights that sum to one. Because $\epsilon = 0.89$, the update fires unless the portfolio falls more than 11% in one period; in the weekly 1999–2024 run it fired on all but 0.4% of its 2,919 updates (learning from 1995). Assets that rose most lose the most weight.

We test four schedules (Table 1).

Table 1: CWMR schedules tested.

Name	Learns from	Trades
Daily	daily ratios	every session
Monthly, daily signal	daily ratios	last session of each month
Monthly bars	month-end to month-end ratios	last session of each month
Weekly bars	week-end to week-end ratios	last session of each ISO week

2.2 Universe and data

The universe is 16 ETFs: SPY, QQQ, IWM, DIA, EFA, EEM and ten Select Sector SPDRs (XLK, XLF, XLY, XLE, XLI, XLP, XLU, XLB, XLRE). EFA and EEM hold non-US stocks and XLRE holds real estate. The ETFs launched between 1993 and 2015, so the universe grows over time: it has 11 members in January 1999 and reaches 16 in October 2015. A fund joins on its first day in the Yahoo data, which for IWM, EFA and EEM is a few sessions after launch. Before that it holds no weight. With fewer than two funds trading the algorithm makes no decision and the account stays in cash. Until a fund trades, the algorithm gives it the average price ratio of the funds that do and splits its weight equally among them, which gives the same portfolio return, so the algorithm runs on the whole universe from the start and keeps what it has learned when a fund joins. The universe is a set of large, liquid ETFs chosen in 2026. Funds that existed in earlier years and later closed or shrank are not in it, which favours any strategy that holds these funds, including the equal-weight benchmark.

Prices are Yahoo daily bars adjusted for dividends and splits, from 1995 so the algorithm has history to learn from before 1999. A strategy decides on a session’s close and fills at the next session’s open, the first price after its decision. Results cover 1999-01-01 to 2024-12-31; 2025 onward is a holdout (Section 7), which used Alpaca minute bars and trades at 15:45 New York time, as the author’s paper-trading system does.

2.3 Costs and benchmarks

Every dollar traded pays a flat cost in basis points (bp), plus the SEC fee on sales (the 2026 rate of \$20.60 per million dollars sold, applied to all years). Turnover is dollars bought plus dollars sold per year, divided by average equity. We measured half-spreads at 15:45 from Alpaca quotes on 45 sessions in 2016–2024, five chosen at random in each year (Table 2).

Table 2: Median half-spreads at 15:45, 2016–2024.

ETF	Half-spread (bp)
SPY	0.17
QQQ, IWM, DIA	0.26–0.30
Most sector SPDRs, EFA	0.41–0.82
EEM, XLRE, XLF	1.20–1.68
Equal-weight average	0.70

We charge 0.7 bp unless stated and report results at 5, 10 and 20 bp. The model has no commissions, no market impact and no taxes; Section 9 discusses each.

The benchmarks are SPY with dividends reinvested and an equal-weight portfolio (EW) of the same ETFs, reset at each month end. EW differs from the constant rebalanced portfolio (CRP)

of Section 3, which resets every day. EW matters more. A strategy that beats SPY only because these ETFs beat SPY has learned nothing.

2.4 Statistics and procedure

We compare Sharpe ratios with the test of Jobson and Korkie (1981) with the correction of Memmel (2003), on daily returns of two strategies over the same days. The test uses their correlation, 0.71 to 0.98 across this study, so it can detect smaller differences than comparing two independent estimates. It assumes independent, normally distributed returns, and these daily returns have fat tails and clustered volatility. We therefore also report a moving-block bootstrap (Künsch, 1989) with 21-day blocks and 10,000 draws, with a 95% interval for the annual Sharpe difference, and show how its p -values change with the block length. The bootstrap is not studentised; Ledoit and Wolf (2008) propose a studentised version for this problem. We give Holm-adjusted p -values for the nine comparisons of Table 5. The tests compare Sharpe ratios of raw returns; the tables report Sharpe ratios in excess of the 3-month Treasury bill rate.

Three tests were pre-registered: each plan was written into the research log and committed to git before the test ran, naming the strategy version, window, settings and pass criterion, and stating that the strategy would not be changed and rerun on that window (Section 8). The texts are in the replication repository with their commit hashes and times. Git commit times come from the author’s machine. For the holdout test, GitHub records the push at 10:06:06 UTC and the backtest database records the run at 10:06:32 UTC. For the 1995–2015 plan (commit `d9a912d`), GitHub records the push at 08:12:12 UTC, before the results commit at 08:12:49 UTC by the author’s clock. For the 1999–2015 plan (commit `97b0d4a`) there is no third-party record; the order rests on commit times set by the author’s machine, public in the btest repository. Everything in Sections 4 to 6 is exploratory. All four schedules were run on 2016–2024 minute data before the 1999–2015 plan was written, so those years are not out of sample for any of them. The 2025 onward holdout had been used by two unrelated single-ETF strategies before this study and by one test in it.

3 Reproducing the original paper

The authors’ repository contains their data, 93 NASDAQ-100 stocks from 1998-01-02 to 2009-12-31. Running their functions reproduces Tables 3 and 4 of Lahanis et al. (2025) exactly: CWMR 1,590.38×, PAMR 788.26×, Anticor 36.89×, FTRL 15.25× and the equal-weight constant rebalanced portfolio (CRP) 12.16×.

The paper describes walk-forward validation of its parameters, but the validation windows lie inside 1998–2009 and its Tables 3 and 4 report the whole sample, so these are in-sample results. Within the sample, mean reversion keeps much of its return under frictions (Table 3).

Four of the paper’s strategies differ from the algorithms they are named after. As implemented, OLMAR and exponential gradient never change their target weights. Both reduce to CRP, and the paper reports identical numbers for all three. CWMR and RMR are modified versions of the published methods. A fifth result depends on the data: the lead of follow the regularized leader (FTRL) over CRP disappears when the 32 tickers with incomplete data are removed. Two bad prints, PCAR in February 2000 and XRAY in July 2004, account for about 30% of CWMR’s final wealth.

Table 3: The authors’ data, 1998–2009. Paired Sharpe test against CRP.

	CAGR	p vs CRP
CRP	23.2%	
CWMR	85.0%	< 0.0001
CWMR, 10 bp	43.6%	0.29
CWMR, one-day delay	45.6%	0.11
PAMR, one-day delay	51.9%	0.0002
PAMR, one-day delay and 5 bp	33.9%	0.49

4 CWMR on the ETFs, 1999–2024

Table 4 runs the four schedules of Table 1 and both benchmarks over 1999–2024 at 0.7 bp per dollar traded. The algorithm learns from 1995 throughout.

Table 4: 1999–2024, at 0.7 bp per dollar traded. CAGR in % a year; Sharpe ratio over T-bills; turnover per year.

	CAGR	Sharpe	Max drawdown	Turnover
Daily	12.13%	0.56	−44.7%	170×
Monthly, daily signal	7.72%	0.38	−53.1%	14×
Monthly bars	10.66%	0.49	−61.4%	14×
Weekly bars	12.10%	0.55	−50.1%	76×
Equal weight (EW)	8.37%	0.42	−54.4%	0.3×
SPY	8.09%	0.40	−55.2%	0×

Daily, weekly-bar and monthly-bar CWMR all return more than both benchmarks; the monthly schedule with a daily signal returns less. Equal weight itself barely beats SPY over the whole period (8.37% against 8.09%).

Table 5: Paired Sharpe ratio tests, 1999–2024: annual Sharpe difference, Jobson–Korkie–Mennel p , block-bootstrap p (21-day blocks) and 95% bootstrap interval.

	Difference	JKM p	Bootstrap p	95% interval
Daily vs EW	+0.13	0.09	0.066	[−0.01, +0.27]
Monthly, daily signal vs EW	−0.05	0.53	0.45	[−0.18, +0.07]
Monthly bars vs EW	+0.05	0.46	0.40	[−0.07, +0.18]
Weekly bars vs EW	+0.12	0.14	0.083	[−0.02, +0.27]
Daily vs SPY	+0.15	0.06	0.043	[+0.01, +0.30]
Monthly, daily signal vs SPY	−0.03	0.73	0.66	[−0.16, +0.10]
Monthly bars vs SPY	+0.07	0.38	0.28	[−0.06, +0.22]
Weekly bars vs SPY	+0.14	0.10	0.031	[+0.02, +0.28]
EW vs SPY	+0.02	0.63	0.55	[−0.04, +0.09]

None of the Sharpe differences against equal weight is significant (Table 5). Against SPY, daily and weekly bars have bootstrap p -values below 0.05 but Jobson–Korkie–Mennel p -values of 0.06 and 0.10. The bootstrap p depends on the block length: the daily return difference between

each variant and equal weight is negatively autocorrelated (lag-1 -0.04 to -0.09), so short blocks overstate its variance and longer blocks give smaller p -values. With 126-day blocks, weekly bars reaches 0.048 against equal weight and 0.012 against SPY. After a Holm correction for the nine comparisons, the smallest p at any block length is 0.11 (weekly bars against SPY, 126-day blocks), and against equal weight it is 0.35.

5 Costs

The flat 0.7 bp is the average half-spread we measured at 15:45 in 2016–2024 (Table 2). These strategies trade at the open, and over years in which we did not measure spreads at all. Table 6 shows each schedule at higher costs, and the cost at which its return falls to equal weight’s or SPY’s.

Table 6: CAGR at higher costs per dollar traded, 1999–2024, and the break-even cost against each benchmark (bp).

	0.7 bp	5 bp	10 bp	20 bp	vs EW	vs SPY
Daily	12.13%	1.56%	−9.47%	−25.09%	2.2	2.3
Monthly, daily signal	7.72%	6.99%	6.16%	4.50%	—	—
Monthly bars	10.66%	10.01%	9.26%	7.78%	16.0	17.9
Weekly bars	12.10%	8.50%	4.46%	−3.17%	5.2	5.5

Daily CWMR’s lead needs costs at the open below about 2 bp per dollar traded for 26 years, and weekly bars below about 5 bp. Monthly bars can bear 16 bp, but its Sharpe ratio is no better than equal weight’s. Estimating historical spreads from daily prices does not work at this scale. We applied the efficient estimator of Ardia et al. (2024), which its authors show improves on the earlier high–low estimators of Corwin and Schultz (2012) and Abdi and Rinaldo (2017), to each ETF’s daily open, high, low and close in each year. On 2016–2024, where half-spreads were measured at 0.17 to 1.68 bp, its yearly estimates range from -34 to $+74$ bp and its medians by ETF from -11 to $+17$ bp; for SPY it gives 11.9 bp against a measured 0.17 bp. Daily price noise swamps spreads this small, so the estimates for earlier years cannot be used. Commissions add to this: weekly bars made 238 to 313 trades a year in 1999–2001, and at the 1999 average commission of the ten largest online brokers, \$15.75 a trade (Unger, 1999), that is roughly a fifth to a quarter of a \$20,000 account each year.

6 The weekly variant

6.1 Robustness to the day the week ends

Weekly bars end on each calendar week’s last session, usually a Friday. We reran the strategy with blocks of 5 sessions at each of the five possible starting offsets instead; holidays make these blocks drift against calendar weeks (Table 7).

Every version returns more than equal weight’s 8.37%, but the Sharpe difference is significant for one of the six before any correction, and the calendar version is the second best. Which day the week ends on moves the return by almost 4 percentage points a year.

The weekly variant is also concentrated. After rebalancing it holds a median of 2 ETFs, and one ETF carries 90% or more of the account in 36% of weeks.

Table 7: Weekly bars by where the week ends, 1999–2024, at 0.7 bp; block-bootstrap p for the Sharpe difference against equal weight.

	CAGR	Sharpe	p vs EW
Calendar weeks (Friday)	12.10%	0.55	0.083
5-session blocks, offset 0	11.01%	0.51	0.30
5-session blocks, offset 1	10.80%	0.49	0.41
5-session blocks, offset 2	13.48%	0.61	0.015
5-session blocks, offset 3	9.64%	0.44	0.86
5-session blocks, offset 4	10.13%	0.47	0.60

6.2 Trading one session late

If the weekly lead came from a reversal that unfolds over days, delaying each trade by one session should cost little. Filling at the open two sessions after the signal instead of one lowers weekly bars from 12.10% to 9.51% a year (Sharpe 0.44), and the bootstrap p against equal weight rises from 0.083 to 0.84. Most of the lead comes from the first session after the signal. Regressing each ETF’s return in excess of SPY’s on its excess return over the signal week gives +0.10 bp per 1% move from the signal’s close to the next open and -3.3 bp from that open to the open after. The reversal is between consecutive opens, which a trader can capture, so the cost of trading at the open, which we have not measured, decides whether the lead was real.

7 The holdout

Before the tests above, we searched for a lower-turnover variant on 2016–2024 minute data, deciding at 15:30 and trading at 15:45: the paper’s 13 strategy families with daily trading, with a daily signal and monthly trades, and with monthly bars, 38 configurations in all. CWMR with a daily signal and monthly trades did best there, 18.3% a year against SPY’s 14.5%, so we pre-registered a test on 2025-01-01 onward. The pass rule was CAGR and Sharpe ratio both above SPY’s. Both came in slightly below, so it failed (Table 8).

Table 8: Pre-registered holdout, 2025-01-01 to 2026-10-02, minute data, trading at 15:45.

	Monthly CWMR	SPY
CAGR	18.28%	18.36%
Sharpe (T-bill)	0.81	0.87
Max drawdown	-20.3%	-18.7%

The paired p was 0.77, and the strategy beat SPY in 8 of 21 months. With 438 sessions the test cannot rule out a small edge either way. Over 1999–2024 the same variant returns 7.72% a year, less than both benchmarks (Table 4): the lead that selected it does not hold over the longer period.

8 Pre-registered tests and deviations

Besides the holdout, two tests were pre-registered. The first asked whether the monthly variant’s lead over SPY appeared over 1995–2015, filling the years before each ETF launched with similar

mutual funds and indexes. It passed, returning 11.57% a year against SPY’s 9.33%, but it decided and filled at the same close (see below). We no longer rely on it: with each ETF’s own history only it returns 6.80% against SPY’s 9.33%.

The second tested weekly and monthly bars on 1999–2015, with CAGR and Sharpe ratio above SPY’s as the pass criterion. Both passed: weekly bars returned 9.82% (Sharpe 0.45) and monthly bars 8.90% (0.40) against SPY’s 4.90% (0.25). Equal weight passes the same criterion (6.59%, 0.33), and we already knew this basket had beaten SPY over that window when we wrote the plan, so the criterion was too easy; equal weight was the right benchmark to name.

Two procedures departed from the plans. The registered designs of both tests decided and filled at each session’s close, so each order filled at the closing price its decision had used, which no one can trade at. This look-ahead error was worth 1.7 percentage points a year to weekly bars on 1999–2015 (11.50% before the correction). We moved fills to the next session’s open and reran the 1999–2015 test with nothing else changed; a later correction to how next-open fills were booked required one further run, which left CAGR and the pass result unchanged. The holdout plan also stated that no strategy had used the holdout window, which was wrong (Section 2), and its pipeline check judged close-to-close fills equivalent to the 15:45 fills, which missed the look-ahead error above.

9 Discussion

The original paper’s results are real, in the sense that its code computes them correctly from its data. On these ETFs, with fills at the next open, the daily and weekly schedules return about 4 percentage points a year more than equal weight at 0.7 bp, but neither lead is statistically significant after correcting for the comparisons, both need costs at the open below a few basis points, and the monthly schedule chosen for the holdout trails both benchmarks.

The weekly gain comes from a one-session reversal between consecutive opens (Section 6.2). Daily close-to-close returns of these ETFs in excess of SPY show negative next-day autocorrelation (median across the ETFs -0.036), which fits bid-ask bounce in closing prices (Roll, 1984); but the strategy’s fills are at the open, so bounce in closing prices does not explain its gain. Two explanations remain. The reversal may be pay for supplying liquidity (Nagel, 2012), earned only by a trader whose costs at the open are low enough; and the variants came out of a search over about 100 configurations across this study (an approximate count), which inflates the best result (Harvey et al., 2016). The smallest unadjusted p in Table 5 at any block length is 0.012 ($z \approx 2.5$); the expected largest of 100 independent standard normal draws is also about 2.5, so a search of this size would often produce it by chance. A deflated Sharpe ratio (Bailey and López de Prado, 2014) would make this precise.

Before calling either schedule an edge we would need:

1. spreads for these ETFs at the open over 1999–2024, measured from historical quote data, to replace the flat cost;
2. a pre-registered test against equal weight, not SPY, on data no variant has touched. The 2025 holdout has been used once; data from 2027 on would be clean;
3. results after commissions for a stated account size, and after tax for taxable accounts. At $76\times$ turnover or more almost every gain is short-term.

10 Conclusion

We reproduced a publicly released set of mean-reversion results exactly, then tested the strongest algorithm on liquid ETFs over 1999–2024. Traded daily or on weekly bars it returns about 4 percentage points a year more than an equal-weight portfolio of the same ETFs at 0.7 bp, but the lead needs costs at the open below about 2 and 5 bp respectively, the weekly lead mostly disappears if trades are made one session later, and no Sharpe ratio difference is significant after correcting for the comparisons. The variant selected on recent data trails both benchmarks over 1999–2024 and failed its pre-registered holdout test. We find no statistically significant edge for this CWMR variant over holding the same ETFs.

Data and code availability

Code that reproduces Section 3, Tables 3 and 4–7, Sections 4 to 6, the results of the 1999–2015 pre-registered test and the autocorrelations from public data, together with the texts of the three pre-registered plans, is at

<https://github.com/yelgabo/cwmr-etf-replication>

It uses the original authors’ public code and data (Lahanis et al., 2025), Yahoo Finance daily prices and the 3-month Treasury bill rate from FRED. The holdout test and the 2016–2024 search of Section 7 used minute bars from Alpaca, which require an account. The backtester, which computes the same tables independently, and the scripts behind those sections are at

<https://github.com/yelgabo/btest>

under docs/research/olps-replication/.

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